MILITARY SPECIFICATION

FREQUENCY METERS, AIRCRAFT 380 TO 420 HERTZ AND 350 TO 450 HERTZ

1. SCOPE

• 1.1 This specification covers the requirements for four types of aircraft frequency meters for operation on 115 volts (V) 400 Hertz (Hz) alternating current (AC) 3 phase, 4-wire "Y" systems.

1.2 Classification. Aircraft frequency meters covered by this specification shall be of the following types, as specified (see 6.2).:

Type A-1	350 to 450 Hz, 2 inch flange-mounted aircraft instrument case.
Type A-2	380 to 420 Hz, 2 inch flange-mounted aircraft instrument case.
Type A-3	350 to 450 Hz, 2 inch clamp-mounted aircraft instrument case.
Type A-4	380 to 420 Hz, 2 inch clamp-mounted aircraft case.

2. APPLICABLE DOCUMENTS

• 2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

FF-N-836	Nut; Square, Hexagon, Cap, Slotted, Castle, Knurled, Welding And Single Ball Seat
FF-S-92	Screws, Machine: Slotted, Cross-Recessed Or Hexagon Head
FF-W-84	Washers, Lock (Spring)
FF-W-92	Washers, Metal, Flat (Plain)
QQ-P-416	Plating, Cadmium (Electrodeposited)
QQ-S-365	Silver Plating, Electrodeposited; General Requirements For

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Military

MIL-M-3171	Magnesium Alloy, Processes For Pretreatment And Prevention Of Corrosion On
MIL-C-5541	Chemical Conversion Coatings On Aluminum And Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series: General Specification For
MIL-A-8625	Anodic Coatings, For Aluminum And Aluminum Alloys
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts: Packaging And Packing Of
MIL-N-25027	Nut, Self-Locking, 250°F, 450°F, And 800°F, 125 KSI FTU, 60 KSI FTU, And 30 KSI FTU
MIL-L-25142	Luminescent Material, Fluorescent
MIL-B-27497	Bearings, Jewel, Sapphire Or Ruby, Sythetic
MIL-G-81704	Glass, Aircraft Instrument, Lighting Wedge And Cover

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking Of U.S. Military Property
MIL-STD-202	Test Methods For Electronic And Electrical Component Parts
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-704	Electric Power, Aircraft, Characteristics And Utilization Of
MIL-STD-810	Environmental Test Methods
MS33558	Numerals And Letters, Aircraft Instrument Dial, Standard Form Of
MS33638	Cases, Instrument, Flange Mounted, Aircraft
MS33639	Cases, Instrument, Clamp-Mounted, Aircraft

PUBLICATIONS

Air Force-Navy Aeronautical Bulletin

ANA Bulletin No. 400	Airborne Electronic And Associated Equipment, Applicable Documents

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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• 2.2 Precedence of documents. In the event of conflict between the requirements of this specification and the applicable documents specified in 2.1, the requirements of this specification shall govern.

3. REQUIREMENTS

• 3.1 Applicability of requirements. Unless otherwise specified, all requirements apply to all types of instruments covered by this specification.

• 3.2 Qualification. Except as specified herein, frequency meters, hereinafter called instrument(s), furnished under this specification, shall be products which are listed or qualified for listing on the applicable qualified products list at the time set for opening of bids (see 6.3). When instruments are required in ranges or scale markings not listed herein, they shall be procured without qualification or first article testing from those suppliers who have qualified instruments with similar ranges and markings.

• 3.2.1 Design changes occuring after qualification approval shall require resubmission for qualification testing, unless a specific waiver is granted by the activity that approved the qualification units.

• 3.3 Parts, materials, and processes. Except as specified herein, all parts, materials, and processes which are not specifically designated herein by reference to specifications and standards and that are necessary for the execution of this specification shall be as specified in AMA Bulletin 400.

a. The requirements of this specification with regard to the use of parts, materials, and processes shall not relieve the contractor of the responsibility for complying with all design and construction requirements and performance requirements set forth herein.

b. When a required part, material, or process is not specifically detailed herein and not covered by AMA Bulletin 400, a nonstandard part, material, or process capable of complying with the requirement may be used, provided the procuring activity specifically approves its use.

• 3.3.1 Metals. Metals shall be of the corrosion-resistant type or shall be suitably protected to resist corrosion during normal service life.

• 3.3.2 Protective treatment. When materials are used in the construction of the instruments that are subject to corrosion in salt air or other atmospheric conditions likely to occur during service usage, they shall be protected against such corrosion in a manner that will in no way prevent compliance with the requirements of this specification. Protective coatings that will crack, chip, or scale with age or extremes of atmospheric conditions shall not be used.

• 3.3.3 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. The use of dissimilar metals separated by suitable nonmetallic insulating material will be permitted.

• 3.3.4 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the meters except where magnetic materials are essential.

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* 3.3.5 Fungus-inert materials. Materials which are nutrients for fungus shall not be used. Fungus inert materials shall be in accordance with requirement 4 of MIL-STD-454.

* 3.3.6 Corrosion fumes. The materials used in the instruments shall not liberate deleterious fumes.

* 3.3.7 Screw threads. The threads of all machine screws 0.060 inch in diameter or larger shall be in accordance with MIL-S-7742.

* 3.3.8 Washers, nuts, and mounting screws. Except for self-locking nuts, washers, nuts, and mounting screws shall conform to FF-W-84, FF-S-92, FF-W-92 and FF-N-836, as applicable. Self-locking nuts shall conform to MIL-N-25027.

* 3.3.9 Bearings and pivots. Moving elements used in the measurement mechanism shall turn on hardened, accurately ground and highly polished steel pivots, bearing on highly polished synthetic sapphire or ruby jewels. The synthetic sapphire or ruby jewels shall be in accordance with MIL-B-27497. Materials other than sapphire and ruby jewels in accordance with MIL-B-27497 may be used provided the instruments are capable of passing all tests specified in Section 4.

* 3.3.10 Glass. The glass used for the face shall be a cover glass for unlighted instruments in accordance with MIL-G-81704.

* 3.3.11 Finishes and protective coatings.

* 3.3.11.1 Aluminum alloy parts. Except as follows, aluminum alloy parts shall be covered with an anodic film conforming to MIL-A-8625.

a. Dials, small holes and case inserts need not be anodized.

b. Aluminum alloys which do not anodize satisfactorily shall be coated with a chemical film in accordance with MIL-C-5541.

c. Where the primary prupose of the treatment is to afford a suitable paint base, chemical treatments in accordance with MIl-C-5541 may be used in lieu of anodizing.

d. Castings containing nonaluminum alloy integral inserts may be treated with a chemical film in accordance with MIl-C-5541 in lieu of anodizing.

e. When abrasion resistance is a factor, chemical films in accordance with MIL-C-5541 shall not be used in lieu of anodizing.

* 3.3.11.2 Magnesium alloy parts. Wherever practicable, magnesium alloy parts shall be surface treated in accordance with MIL-M-3171 to provide protection against corrosion.

* 3.3.11.3 Cadmium plating. Cadmium plating shall be in accordance with QQ-P-416.

* 3.3.11.4 Fluorescent-luminescent markings. Fluorescent-luminescent marking material shall be in accordance with type II of MIL-L-25142.

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• 3.3.11.5 Colors. Lusterless black and white colors shall be in accordance with the 37000 series of FED-STD-595.

b. 1.6 Silver plating. Silver plating shall be in accordance with type II, Grade A of QQ-S-365.

• 3.4 Design and construction. Unless otherwise specified herein, the instruments shall be designed and constructed:

a. For operation from the normal, abnormal, and emergency conditions specified in MIL-STD-704; 115/200 volts, 400 Hz, 3 Phase, 4-wire "Y" systems, equipment categories A, B, and C (nominal of 115 volts, 400 Hz line to neutral.) The instruments shall be delivered complete, tested, and ready for installation.

b. To withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

• 3.4.1 Environmental. The instruments shall be designed and constructed to operate under any one of and all possible combinations of the following environmental conditions:

a. Shock, 50 gravity (g) units. When tested as specified in 4.7.12, permanent errors in indication shall be not greater than ±2 percent of applied frequency. No part shall become loosened or unduly damaged so as to affect the operation of the instrument.

b. Vibration, 20g. When tested as specified in 4.7.13, permanent errors in indication shall be not greater than ±1 percent of the applied frequency. No part shall become loosened or unduly damaged so as to affect the operation of the instrument.

NOTE: The permanent error in indication due to shock and vibration is not cumulative, that is, permanent error due to shock and vibration shall be not greater than ±2 percent of the applied frequency.

c. Temperature.

(1) Operating. -54° Centigrade (C) to +71°C.

(2) Nonoperating. -62° to +85°C.

d. Relative humidity (rh). Up to 100 percent.

e. Fungus. Moist fungus encountered in tropical and subtropical climates.

f. Sand and dust. Exposure to airborne sand and dust particles.

g. Explosive atmosphere.

h. Salt spray. Exposure to atmosphere containing salt laden moisture.

i. Altitude. Pressure ranging from sea level to 70,000 feet.

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* 3.4.1.1 Instruments delivered under this specification are not repairable. The only adjustment permitted is the pointer zero adjustment (see 3.4.6.4).

* 3.4.2 Operating life. The instruments shall have an operating life of 50,000 ON-OFF cycles (see Table II, Group A and 4.7.23).

* 3.4.3 Magnetic shielding. The instruments shall be magnetically shielded so that it shall not cause a compass deflection of more than 5 degrees when tested as specified in 4.7.20.

* 3.4.4 Dielectric withstanding voltage. The instruments shall be capable of withstanding a voltage of 2500 volts root mean square (rms) AC, at commercial frequency, applied for one minute at sea level.

3.4.5 Case.

* 3.4.5.1 Dimensions.

a. Types A-1 and A-2. Except for the depth, the type A-1 and A-2 case shall be in accordance with the two inch case requirements of MS33638. The depth (detail L of MS33638) shall be not greater than 4 inches. The maximum depth of the case, with terminal studs, shall be not greater than 4.656 inches.

b. Types A-3 and A-4. Except for the depth, the case shall be in accordance with the two inch case requirements of MS33639. The depth (detail L of MS33639) shall be not greater than 4 inches. The maximum depth of the case, with terminal studs, shall be not greater than 4.656 inches.

* 3.4.5.2 Material.

* 3.4.5.2.1 Body. The case shall be made of metal. The case shall be uniform in texture and shall have a smooth surface. The case shall be finished in a durable lusterless black.

* 3.4.5.2.2 Cover glass. The cover glass shall be clear and free from flaws which interfere with normal reading of the instrument. The glass shall not fog when subjected to the environmental tests specified in Section 4.

* 3.4.5.2.3 Terminal studs. The terminal studs shall be No. 10-32 and shall be provided with four nuts, two flat washers and two spring type lock washers. When tested as specified in 4.7.15 and 4.7.16 the terminals, case and instrument shall not be damaged. At least 0.60 inches of the terminal studs shall be available to accept the aircraft wiring lug and the above specified nuts and washers.

* 3.4.5.3 Pull strength of mounting lugs. Mounting lugs of the completely assembled case shall withstand, without fracture, a load of 175 pounds applied for one minute along the nut axis and toward the front of the instrument.

* 3.4.5.4 Mounting screws. The contractor shall furnish sufficient mounting screws for installation of Type I and II instruments on the rear of a panel 3/16 inch thick. They shall be No. 6-32 NC-2A round head brass machine screws, 1 inch long, having a black oxide finish.

* 3.4.5.5 Envelope. A transparent envelope containing the terminal nuts and washers, mounting screws, range marking decals, and installation instructions shall be provided with each instrument.

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* 3.4.5.6 Seal. The instruments shall be hermetically sealed and filled with a mixture of 88 to 92 percent nitrogen and 8 to 12 percent of helium. The mixture shall be at least 98 percent pure, be free of dust particles, and contain not more than 0.006 milligram of water vapor per liter (dew point -65°C) at the filling pressure. The pressure of the filling mixture, in the instrument's case, shall be approximately one atmosphere (sea level ±250 feet).

* 3.4.6 Pointer. A lance type pointer may be used.

* 3.4.6.1 Pointer dimensions.

a. The width of the pointer tip shall not be wider than the width of a minor graduation on the scale.

b. Pointer length. The length of the pointer shall be such that the tip of the tapered end shall overlap not more than 50 ±5 percent of the shortest graduation.

* 3.4.6.2 Pointer clearance. To avoid reading errors caused by parallax, the clearance between the pointer and the graduated scale shall be not greater than 0.050 inches.

* 3.4.6.3 Pointer marking. The entire visible portion of the pointer shall be marked with fluorescent-luminescent material.

* 3.4.6.4 Pointer zero correction. A pointer zero correction control, accessible at the front of the case, shall be provided.

* 3.4.6.4.1 Adjustment range. The pointer zero correction control shall provide an adjustment range of at least ±3 degrees around the zero mark on the scale. The control shall be capable of being rotated 360 degrees without damage to the instrument. The control shall be insulated to eliminate all danger to operating personnel.

* 3.4.6.4.2 Access to pointer zero correction control (Types A-1 and A-2 only). To deny access to an adjustment by unauthorized personnel, a cover plate shall be provided on the Type A-1 and A-2 instruments. The instruments shall be delivered with the cover plate installed.

* 3.4.6.4.3 Meter zero. The meter zero shall be at 350 Hz or below or at 380 Hz or below as required by the type designation. The zero (0) shall be marked on the scale.

3.4.7 Scale and dial.

3.4.7.1 Scale.

* 3.4.7.1.1 Type A-1. The scale of the type A-1 instrument shall be substantially as shown in Figure 1. The range of the scale and resolution shall be exactly as shown in Figure 1.

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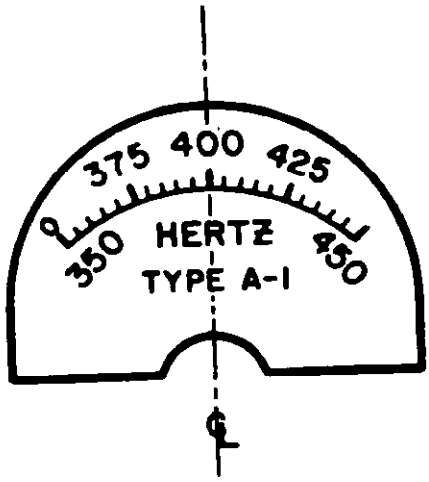


FIGURE 1.

FREQUENCY METER SCALE

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• 3.4.7.1.2 Type A-2. Except for the range, resolution, and type designation, the scale of the type A-2 instrument shall be substantially as shown in Figure 1. The range shall be from 380 to 420 Hz. Numerical markings shall be provided at 380, 390, 400, 410, and 420 Hz. The scale shall be graduated in increments of 2 Hz.

• 3.4.7.1.3 Type A-3. Except for the type designation, the scale of the type A-3 instrument shall be as specified in 3.4.7.1.1.

• 3.4.7.1.4 Type A-4. Except for the type designation, the scale of the type A-4 instrument shall be as specified in 3.4.7.1.2.

• 3.4.7.1.5 Scale length. The scale length shall be not less than 1.5 inches as measured by the arc described by the pointer tip.

• 3.4.7.1.6 Scale markings. The dimensions of the scale and dial markings shall be as specified in Table I. The scale shall be marked with fluorescent-luminescent material. The style of the markings shall be in accordance with MS33558. Numerals shall distinctly indicate the graduation to which each applies. Any confusion resulting in doubt as to which graduation the numeral applies will be cause for rejection. When several numerals are used in one group, the space between numerals shall be not less than 1/64 inch.

TABLE I - SCALE AND DIAL MARKINGS

	Height or Length Inch ±1/64	Width of Line or Graduation Inch ±0.005
Group A: Fluorescent-luminescent markings		
All Numerals	7/64	0.030
Major Graduations	5/32	0.030
Lettering "HERTZ"	5/32	0.030
Section Line	----	0.015
Group B: Lusterless White		
Minor Graduations	7/64	0.020
Lettering "Type A-1"	1/16	0.015
Meter Zero "0"	1/16	0.015

• 3.4.7.1.7 Scale readability. The scale and transparent front shall be designed so that the full scale of the instrument can be easily read at any angle up to 45 degrees below and 30 degrees on both sides of the axis through the center of the instrument, perpendicular to the plane of mounting, with illumination coming from above and in front and at an angle of 45 degrees from the same axis. Under these conditions, there shall be no objectionable shadows on the graduated scale.

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* 3.4.7.2 Dial.

* 3.4.7.2.1 Dial plates. Dial plates shall be constructed of aluminum or aluminum alloy, capable of withstanding the tests specified herein. The dial plates shall be securely fastened to the frame of the mechanism. If screws are used to attach the dial plate, they shall be so located that they will not interfere with the proper location of any dial or scale marking.

* 3.4.7.2.2 Dial color. The dial color shall be lusterless black.

* 3.4.7.3 Range marking decals. Red, green, and yellow adhesive backed decals, to permit range marking the face of the cover glass, shall be provided. Each decal shall be 0.0625 ±0.01 inches wide and at least 1 inch long.

* 3.4.8 Weight. The weight of the instrument, with mounting hardware, shall be not greater than 1 pound.

* 3.5 Performance requirements.

* 3.5.1 Accuracy. Unless otherwise specified herein (see 3.4.1) all changes in accuracy are temporary.

* 3.5.1.1 Rated accuracy. When operated at voltages from 98 to 132 volts rms and temperatures from -54° to +71°C, the accuracy shall be within ±1.5 percent of the applied frequency.

* 3.5.1.1.1 Effect of harmonic distortion on accuracy. The rated accuracy shall not change more than:

a. ±0.5 percent with up to 80 percent of 2nd, 3rd, and 4th harmonic distortion present.

b. ±0.5 percent with up to 70 percent of 5th, 6th, and 7th harmonic distortion present.

c. ±1 percent with up to 60 percent of 8th and 9th harmonic distortion present.

d. ±1 percent with up to 50 percent of 10th and 11th harmonic distortion present.

e. ±1 percent with up to 25 percent of 12th through 20th harmonic distortion present.

f. ±1 percent with up to 20 percent of 21st through 30th harmonic distortion present.

g. ±0.75 percent with up to 15 percent of 31st through 100th harmonic distortion present.

* 3.5.1.2 Magnetic panel influence. The change in accuracy when operated on a magnetic and nonmagnetic panel shall be not greater than ±1 percent of the scale range.

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• 3.5.2 Position influence. When the frequency meter is tilted 60 degrees from the normal vertical position in any direction, while energized to indicate 352, or 382 Hz, depending on type, the pointer tip shall remain within ±1 percent of the scale range of the initial position.

• 3.5.3 Voltampere (VA) consumption. The va consumption of the frequency meter shall be not greater than 3VA from 350 to 450 Hz.

• 3.5.4 Damping factor. When measured as specified in 4.7.9, the damping factor shall be at least 1.5.

• 3.5.5 Response time. When measured as specified in 4.7.10, the response time shall be not greater than 2 seconds.

• 3.5.6 Friction error. When measured as specified in 4.7.11, the friction error shall be not greater than ±1 percent.

• 3.6 Identification of product. The following data shall be marked in accordance with MIL-STD-130 on the side(s) of the case body:

a. Frequency Meter, AC, insert the frequency range of the type being procured, 115V, Type A-(insert type number) for 115/200V 3 phase 400 Hz 4 wire "Y" System.

b. For use on 115V - 120V (line to neutral connection). DO NOT USE ON 200V (LINE TO LINE CONNECTION).

c. Specification MIL-F-8378.

d. Manufacturer's Part No. _____.

e. Contract Number.

f. National Stock Number.

• 3.7 Workmanship. The workmanship shall be such that no evidence of any physical damage or defect shall be evident upon examination of assembly and fit, materials, parts, freedom from binding, wiring assembly (lacing) and insulation, soldering, impregnation of coils, markings and identification of parts, freedom of parts from burrs and sharp edges, finishes, and external alignment of component parts and assemblies.

4. QUALITY ASSURANCE PROVISIONS

• 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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* 4.2 Classification of inspection. Examination, sampling, and testing of the instruments shall be classified as follows:

* 4.3 Test conditions. Unless otherwise specified, the instruments shall be tested under the following conditions:

a. Temperature - Room ambient, 25°C, ±15°C.

b. Altitude - Normal ground.

c. Relative humidity - Room ambient up to 90 percent rh.

d. Power source - 115 volts *rms*, ±0.1 percent; 400 Hz, ±0.1 percent.

* 4.3.1 Mounting. Unless otherwise specified herein, inspection performed on instruments, including all inspections requiring scale readings, shall be performed with the instrument mounted on a vertical panel and in the position in which it is normally used.

* 4.3.2 Number of meter readings. Unless otherwise specified herein, a minimum of six meter readings shall be taken at approximately equidistant marked points on the scale, including the zero and end-scale points.

* 4.3.3 Tapping. When taking meter readings, either the meter or its support shall be tapped lightly, as with the finger, except when determining the friction error.

* 4.3.4 Parallax. Care should be exercised to avoid the effect of parallax when taking meter readings.

* 4.3.5 Thermal stability. Unless otherwise specified herein, meters shall be maintained at the specified inspection temperature for a time sufficient to attain thermal stability prior to reading the meters. A period of not less than two hours is generally required for this purpose.

* 4.3.6 Adjustments during testing. No adjustments to the meters shall be permitted during the test.

* 4.3.7 Reference standards. Reference standards shall be used to check the indications of meters submitted for inspection. All test voltages and test frequencies shall be within ±0.1 percent.

* 4.3.8 Method of calibration. In cases of dispute, but not limited thereto, the pointer of the meter under inspection shall be set to the center of the scale marking at which the calibrations are being made and the value measured by the reference standard.

* 4.3.9 Rate of temperature change. During temperature tests, the rate of temperature change shall be not less than 2°C per minute and not more than 10°C per minute.

* 4.4 Quality conformance inspection. Quality conformance inspection shall consist of:

a. Individual tests (4.4.1).

b. Sampling tests (4.4.2).

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* 4.4.1 Individual tests. After the instruments have been properly calibrated and assembled, each instrument shall be subjected to the individual tests specified in Table II, Groups A and B. The tests need not be performed in the sequence listed in Table II.

* 4.4.2 Sampling tests.

a. Nine instruments selected at random from each production lot of 200 or less serially numbered instruments shall be subjected to the tests specified in Table III, Groups A, B, and D. Group C shall be performed when specified in the contract or order (see 6.2). After being reworked as necessary and passing the tests specified in 4.4.1, these instruments may be delivered as part of the contract quantity.

b. Two disassembled instruments shall be subjected to the tests specified in Table III, Group E. After final assembly, these instruments may be delivered as part of the contract quantity with acceptance being as specified in 4.4.1.

* 4.4.2.1 Test sequence. Unless otherwise specifed in the test procedure(s), the tests shall be performed in the sequence specified in Table III.

* 4.4.2.2 If the number of failures exceeds the total permitted in Table III, a second lot of samples shall be selected and tested. If any of the samples in the second lot fail any of the tests specified in Table III, the entire production lot that the samples represent shall then be subjected to the test(s) that the samples failed. Production units subjected to and passing the test(s) that the samples failed may, after reconditioning as necessary at no cost to the Government, be offered to the Government for acceptance.

* 4.5 Qualification inspection.

* 4.5.1 Classification of qualification tests. Qualification tests shall consist of:

a. Visual and mechanical.

b. Performance.

c. Environmental.

d. Operating life.

4.5.2 Qualification test samples.

4.5.2.1 Number of samples. The qualification test samples shall consist of twelve instruments complete with schematic diagram, installation instructions, terminal nuts and washers, mounting hardware, and range marking decals, and two disassembled instruments. Prior to submission for qualification testing:

a. Six of the qualification test samples shall have been subjected to and passed the tests specified in Table III, Groups A, B, C, and D, and the two disassembled instruments shall have been inspected as specified in Table III, Group E. All instruments that have been tested as specified in Table III shall be so identified.

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TABLE II - QUALITY CONFORMANCE, INDIVIDUAL TESTS

Examination or Test	Requirement	Test Method	Number of Failures Permitted
Group A - Visual and Mechanical			
Screw threads	3.3.7	4.7.1	0
Washers, nuts and mounting screws	3.3.8	4.7.1	0
Body	3.4.5.2.1 and 3.4.11.5	4.7.1	0
Cover glass	3.4.5.2.2 and 3.3.10	4.7.1	0
Envelope	3.4.5.2.3, 3.4.5.4, 3.4.5.5 and 3.4.7.3	4.7.1	0
Pointer dimensions	3.4.6.1	4.7.1	0
Pointer marking	3.4.6.3	4.7.1	0
Access to pointer zero correction control	3.4.6.4.2	4.7.1	0
Scale markings	3.3.11.4, 3.3.11.5, 3.4.6.4.3 and 3.4.7.1 through 3.4.7.1.4	4.7.1	0
Dial color	3.3.11.5 and 3.4.7.2.2	4.7.1	0
Identification of product	3.6	4.7.1	0
Workmanship	3.7	4.7.1	0
Group B - Performance			
Position influence	3.5.2	4.7.8	0
Damping factor	3.5.4	4.7.9	0
Response time	3.5.5	4.7.10	0
Rated accuracy	3.5.1.1	4.7.2	0
Friction error	4.5.6	4.7.11	0

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TABLE III - SAMPLING AND QUALIFICATION TESTS

Examination or Test	Requirement	Test Method	Number of Failures Permitted
Group A - Visual and Mechanical:	All tests specified in Table II, Group A and:		
Dimensions	3.4.5.1	4.7.1	0
Seal	3.4.5.6	4.7.1	0
Scale readability	3.4.7.1.7	4.7.1	0
Weight	3.4.8	4.7.1	0
Group B - Performance:	All tests specified in Table II, Group B and:		
Magnetic panel influence	3.5.1.2	4.7.3	0
Effect of harmonic distortion on accuracy	3.5.1.1.1	4.7.4	0
Volt - ampere consumption	3.5.3	4.7.6	0
Group C - Operating life	3.4.2	4.7.23	1
Group D - Environmental			
Shock	3.4.1a	4.7.12	1
Vibration	3.4.1b	4.7.13	1
Temperature	3.4.1c	4.7.5	0
Relative humidity Sand and dust, explosive atmosphere, and altitude	3.4.1d, f, g and i	4.7.14	0
Fungus	3.4.1e	4.7.21	0

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TABLE III - SAMPLING AND QUALIFICATION TESTS (Cont'd)

Examination or Test	Requirement	Test Method	Number of Failures Permitted
Salt spray	3.4.1h	4.7.18	0
Magnetic Shielding	3.4.3	4.7.20	0
Dielectric withstanding voltage	3.4.4	4.7.19	0
Terminal studs	3.4.5.2.3	4.7.15 and 4.7.16	0
Pull strength	3.4.5.3	4.7.17	0
Seal	3.4.5.6	4.7.22	0
Group E - Visual and Mechanical (disassembled, 2 instruments)		As specified in Table II, Group A and Table III, Group A and:	
Protective Treatment	3.3.2	4.7.1	0
Dissimilar metals	3.3.3	4.7.1	0
Nonmagnetic materials	3.3.4	4.7.1	0
Bearings and pivots	3.3.9	4.7.1	0
Aluminum alloy parts	3.3.11.1	4.7.1	0
Magnesium alloy parts	3.3.11.2	4.7.1	0
Cadmium plating	3.3.11.3	4.7.1	0
Silver plating	3.3.11.6	4.7.1	0
Pointer clearance	3.4.6.2	4.7.1	0
Dial plate	3.4.7.2.1	4.7.1	0

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b. Six of the qualification test samples shall have been subjected to and passed the tests specified in Table II, Groups A and B.

4.5.2.1.1 Multiple submission. For multiple submission for qualification testing:

a. Types A-1 and A-3. The test samples shall consist of six type A-1 and type A-3 instruments complete with the data and hardware specified in 3.6.2.1 and two disassembled instruments (one type A-1 and one type A-3).

(1) Three of the type A-1 and type A-3 and the two disassembled instruments shall have been tested as specified in 4.5.2.1a.

(2) Three of the type A-1 and type A-3 instruments shall have been tested as specified in 4.5.2.1b.

(3) When six instruments are required for testing in 4.7, they shall be three type A-1 and type A-3 instruments.

b. Types A-2 and A-4. Multiple submission for qualification testing of types A-2 and A-4 shall be as specified in 4.2.5.1.1a for types A-1 and A-3.

4.5.2.2 Test report. The manufacturer shall forward a test report, certified by a registered professional or chief engineer, to the Government with the application, including records of the tests for each unit submitted. These records shall show actual readings taken wherever applicable, rather than check marks indicating conformance.

4.5.3 Discontinuance of tests. The Government maintains the prerogative of stopping the qualification testing at any time a test sample(s) fails to meet the specification requirements. Return of the test sample(s) to the manufacturer may be required whenever the Government determines the sample(s) has not complied with the specification requirements.

• 4.6 Equipment required. The following items of test equipment, or units of equal or greater accuracy and range, shall be used to perform the tests (see 4.7). The test equipment shall conform to laboratory standards whose calibration is traceable to the prime standards at the National Bureau of Standards (NBS).

Equipment	Model
Variable frequency power supply	Any type capable of providing: 50 volt amperes, 98 to 132 volts, 350 to 450 Hz.
Stop watch	Any type capable or measuring the response time to within ±0.01 second.
Environmental test equipment	As specified in MIL-STD-202, for the test being performed and as required to perform the tests specified herein.
Timer	Any type capable of switching the instruments ON/OFF at the rate specified in 4.7.23f.
Frequency counter	Any type capable of measuring and displaying 350 to 450 Hz to within ±0.1 percent.

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True RMS digital voltmeter	Any type capable of measuring to within ±0.1 percent of indicated value and displaying the voltage to the resolution specified in Table IV.
Signal generators	Any type capable of providing the 400 Hz to 40 kilohertz (kHz) specified in 4.7.4.
Amplifier	Any type capable of amplifying the 400 Hz to 40 kHz signals specified in 4.7.4.

* 4.7 Test methods. The following tests shall be performed.

* 4.7.1 Visual and mechanical. The instruments shall be examined to verify compliance with the specified requirements.

* 4.7.2 Rated accuracy.

a. Types A-1 and A-3:

(1) Measure and record the accuracy at 350, 375, 400, 425, and 450 Hz at 115 volts rms. The accuracy shall be within ±1.5 percent of the applied frequency.

(2) Repeat step (1) at 98 and 132 volts rms.

b. Type A-2 and A-4:

(1) Measure and record the accuracy at 380, 390, 400, 410, and 420 Hz at 115 volts rms. The accuracy shall be within ±1.5 percent of the applied frequency.

(2) Repeat step (1) at 98 and 132 volts rms.

* 4.7.3 Magnetic panel influence. Performed on a magnetic panel, 4.7.2 shall be repeated for the magnetic panel influence test. The accuracy shall change not more than ±1 percent at the applied frequency.

* 4.7.4 Effect of harmonic distortion on accuracy.

a. Connect the equipment as shown in Figure 2.

b. With the harmonic signal generator off, adjust the fundamental signal generator for a 400 Hz output at 93.70 volts as indicated on the True RMS digital voltmeter.

c. With the fundamental signal generator off, adjust the harmonic signal generator for a 800 Hz output at 74.90 volts as indicated on True RMS digital voltmeter.

d. Turn the fundamental signal generator on. The True RMS digital voltmeter should indicate 120 volts.

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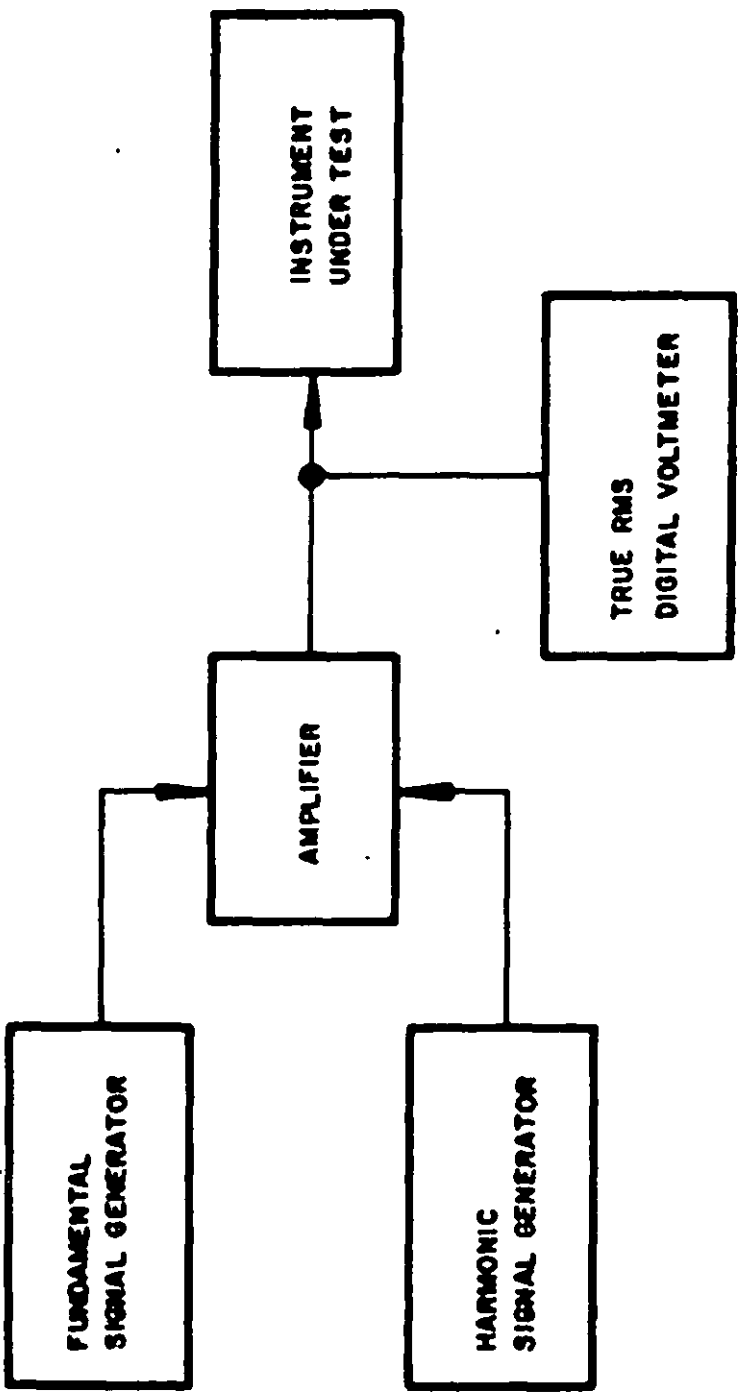


FIGURE 2. HARMONIC DISTORTION TEST SET UP

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(1) Do not exceed 120 volts.

(2) Vary the phase of the harmonic signal generator for worst case harmonic distortion. Repeat steps b, c, and d, as necessary.

e. The accuracy of the instrument under test shall be within the limits specified in Table IV.

f. Repeat the above procedure for all harmonics listed in Table IV.

TABLE IV - HARMONIC DISTORTION LEVELS AND ACCURACY

Harmonic	Output Voltage Fundamental Oscillator	Output Voltage Harmonic Oscillator	Percent Harmonic Distortion	Fundamental	Accuracy Change % of Fundamental
2nd	93.70	74.90	80	400 Hz	±0.5
5th	98.30	68.80	70	400 Hz	±0.5
8th	102.90	61.70	60	400 Hz	±1.0
10th	107.30	53.60	50	400 Hz	±1.0
20th	116.50	29.10	25	400 Hz	±1.0
30th	117.70	23.40	20	400 Hz	±1.0
40th	118.70	17.70	15	400 Hz	±0.75
50th	118.70	17.70	15	400 Hz	±0.75
60th	118.70	17.70	15	400 Hz	±0.75
80th	118.70	17.70	15	400 Hz	±0.75
100th	118.70	17.70	15	400 Hz	±0.75

• 4.7.5 Operating and storage temperature. Throughout this test, there shall be no evidence of moisture (fog or frost) within the instruments. The relative humidity need not be controlled during this test.

a. Place the instruments in a test chamber and repeat 4.7.2a or b, as applicable, to the type under test.

b. With the instruments operating, increase the temperature to +55°C. Maintain +55°C for two hours.

c. Repeat 4.7.2a(1) or b(1) as applicable to the type under test.

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d. With the instruments operating, increase the temperature to +71°C; maintain from +69° to +71°C for four hours.

e. Repeat step c above.

f. With the instruments *nonoperating*, increase the temperature to +85°C; maintain +83° to +85°C for 72 hours.

g. With the instruments nonoperating, cycle the temperature with the rh uncontrolled as follows:

(1) From +85°C to -62°C within 15 minutes; maintain -60° to -62°C for 1/2 hour. There shall be no evidence of fog or frost within the instrument.

(2) From -62°C to +85°C within 15 minutes; maintain +83° to -85°C for 1/2 hour.

(3) From +85°C to -62°C within 15 minutes; maintain -60° to -62°C for 48 hours. There shall be no evidence of fog or frost within the instrument.

h. Increase the temperature to -54°C; maintain -52° to -54°C for four hours. There shall be *no evidence of fog or frost* within the instrument.

i. Repeat step c above.

j. With the instruments operating, increase the temperature to 0°C; maintain 0°C, -0° +2°, for two hours.

k. Repeat step c above.

l. With the instruments operating, increase the temperature to +23°C; maintain +23° ±2°C for two hours.

m. Repeat 4.7.2a or b as applicable to the type under test.

4.7.6 VA consumption. To be performed after all other electrical tests are completed. The VA consumption shall be measured and recorded in terms of VA at 350, 400, and 450 Hz at 115 volts rms or 380, 400, and 420 Hz at 115 rms as applicable to the type under test. The VA consumption shall be not greater than 3 VA.

• 4.7.7 Pointer zero correction. To be performed after all electrical tests are completed.

a. Verify the access to the pointer zero correction control complies with 3.4.6.4.

b. With the instrument unenergized and in a vertical position, rotate the pointer zero correction control 360 degrees.

c. Record the adjustment range.

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d. The pointer shall rotate 360 degrees without damage to the instrument and shall provide an adjustment range of at least ±3 degrees around the zero mark on the scale.

e. Adjust the pointer zero correction control until the pointer is exactly at the zero point.

f. Replace the pointer zero correction cover plate.

* 4.7.8 Position influence.

a. Place the instruments in a vertical mounting position and apply 355 Hz or 382 Hz (depending on the instrument type under test) at 115 volts rms.

b. Tilt the instruments 60 degrees up.

c. Observe and record the pointer deflection.

d. Return to the vertical mounting position.

e. Tilt the instruments 60 degrees down and repeat 4.7.8c and d.

f. Tilt the instruments 60 degrees left and repeat 4.7.8c and d.

g. Tilt the instruments 60 degrees right and repeat 4.7.8c and d.

h. The position influence shall be not greater than that specified in 3.5.2.

* 4.7.9 Damping factor.

a. Electric power sufficient to produce momentary end scale deflection shall be applied to the instrument and the maximum momentary deflection observed and recorded.

b. After the pointer has come to rest, the steady deflection shall be observed and recorded

c. The damping factor shall be taken as the ratio of the steady deflection, in angular degrees, to the difference between the maximum momentary deflection and the steady deflection, in angular degrees; for example, if the maximum momentary deflection is 90 degrees and the steady deflection is 75 degrees, the difference between the two is 15 degrees, and the damping factor (the ratio of 75 degrees to 15 degrees) is 5.

d. The damping factor shall be at least 1.5.

* 4.7.10 Response time.

a. Steady electric power sufficient to produce momentary end-scale deflection shall be applied to the meter and the length of time, in seconds, required for the pointer to come to apparent rest, shall be measured and recorded. (The pointer shall be considered as having come to apparent rest when it has attained the actual point of rest within 1 percent of full scale value.)

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b. The test shall be repeated five times, and the average length of time recorded for all measurements shall be considered as the response time of the meter.

c. The response time shall be not greater than 2 seconds.

• 4.7.11 Friction error.

a. Friction error shall be determined by taking and recording ascending untapped readings at all numerical scale graduations and untapped descending readings at all numerical scale graduations (see 4.3.3).

g. The difference between the corresponding ascending and descending readings, as indicated on a reference standard, shall be not greater than ±1 percent.

• 4.7.12 Shock. The instruments shall be tested in accordance with Method 213 of MIL-STD-202 and the following:

a. Test condition. A

b. Mounting fixture. Except for the mounting holes, the meter shall be securely mounted in a vertical position on a shock test panel similar to that shown in Figure 3. The mounting holes in the shock test panel shall be cut and drilled to fit the dimensions of the instruments being tested.

c. Electrical load conditions. All of the instruments shall be operated.

(1) Two at 132V rms, 350 Hz.

(2) Two at 98V rms, 400 Hz.

(3) Two at 115V rms, 450 Hz.

d. The following shall be used in lieu of 4.1 Method 213 of MIL-STD-202.

(1) The instruments shall be mounted in the normal vertical position and subjected to 10 shocks of 50g.

(2) The instruments (or shock test panel) shall then be rotated 90 degrees in the plane of mounting and subjected to 10 shocks of 50g.

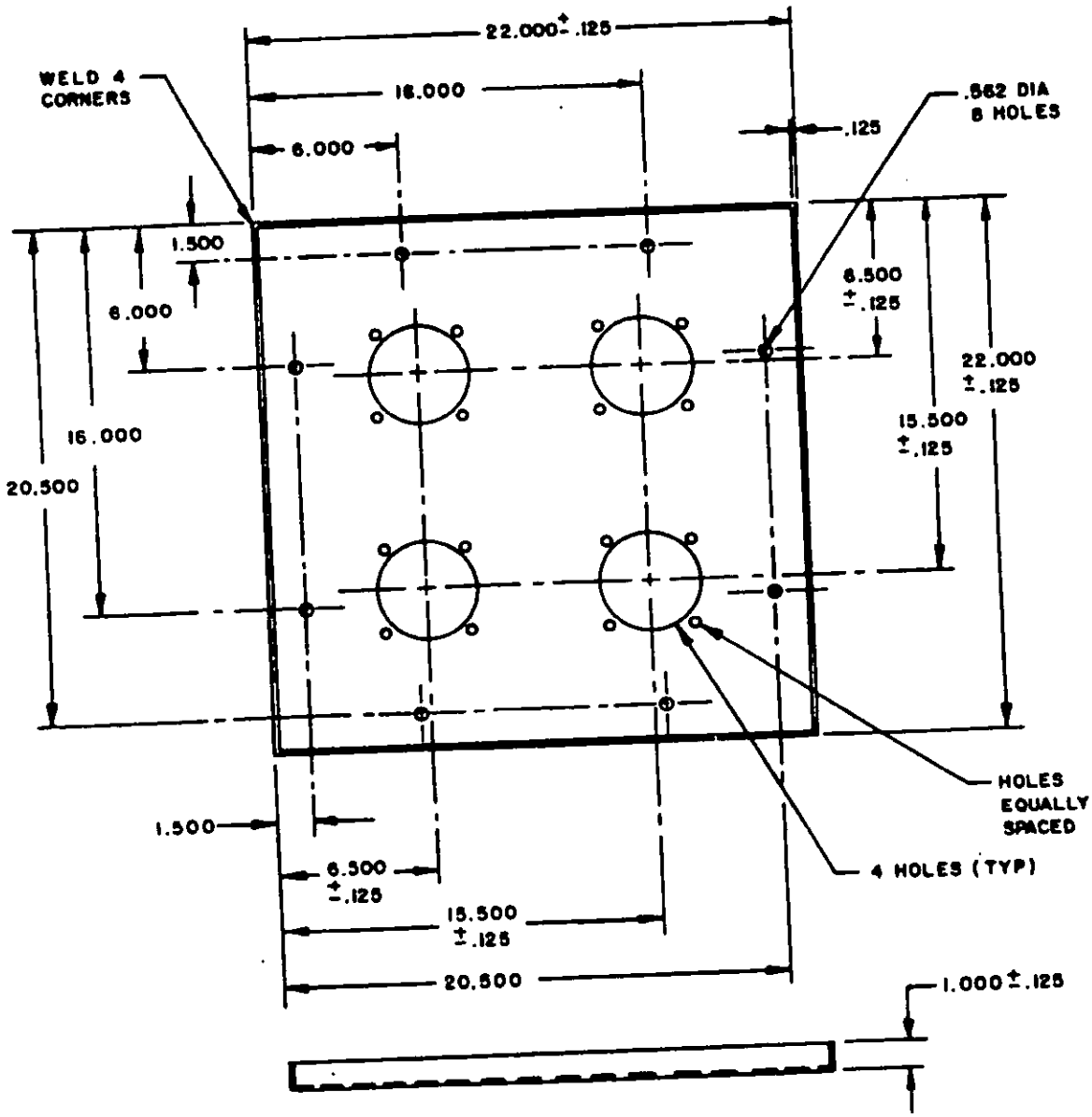
(3) The instruments (or shock test panel) shall then be mounted in a horizontal plane (face up) and subjected to 10 shocks of 50g.

(4) The instruments (or shock test panel) shall then be rotated 90 degrees in a horizontal plane (face up) and subjected to 10 shocks of 50g.

e. Measurements during and after tests.

(1) After each blow the panel and instrument mounting hardware shall be examined and tightened as necessary. Loosening of instrument mounting hardware and terminal nuts shall be considered a failure.

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NOTES:
1. ALL DIMENSIONS IN INCHES.
2. UNLESS OTHERWISE SPECIFIED, TOLERANCES ARE ± .005 ON DECIMALS, AND ± 1/2° ON ANGLES.
3. MATERIAL SHALL BE STEEL.

FIGURE 3.
EXAMPLE OF SHOCK – TEST PANEL

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(2) Before and after the shock test, perform the tests specified in 4.7.2a and b above. Permanent errors in indication as a result of the shock test shall be not greater than ±2 percent of the applied frequency. Also, there shall be no other damage to the instrument.

* 4.7.13 Vibration. The meters shall be tested in accordance with Method 204 of MIL-STD-202 and the following:

a. Test condition A except a level of 20g shall be used.

b. Mounting. Rigidly attached to the vibration test machine.

c. Electrical load condition. Three of the six instruments shall be operating at 400 Hz, 115 volt rms.

d. Measurements after test: As specified in 4.2.1a.

e. Permanent errors in indication due to vibration shall be not greater than ±1 percent of the applied frequency. (See NOTE under 3.4.1).

* 4.7.14 Humidity, explosive atmosphere, sand and dust, and altitude. Because the meters are hermetically sealed, separate humidity, explosive atmosphere, sand and dust, and altitude tests are not required.

* 4.7.15 Pull strength of terminals and case. The pull strength of the terminals shall be tested in accordance with Method 211 of MIL-STD-202 and the following:

a. Test condition. A

b. Method of holding. The case shall be firmly held and the pull applied separately to each terminal.

c. Applied force. 20 pounds. The terminals, case, and instrument shall not be damaged.

d. Measurements after test. Perform 4.7.19.

* 4.7.16 Turn strength of terminals. The turn strength of the terminals shall be tested in accordance with Method 211 of MIL-STD-202 and the following:

a. Test condition. E

b. The force shall be applied to each terminal nut. The terminals shall not turn and the case and instrument shall not be damaged.

c. Measurements after test. Perform 4.7.19.

* 4.7.17 Pull strength of case. To be performed after the pull strength of terminals and case test. Two instruments shall be tested.

a. The instruments shall be mounted face downward on the movable head of a suitable testing machine with the diameter of the case in a horizontal plane so that the mounting lugs receive no added support. A suitable pin shall be inserted through the mounting hole and attached to the pull strap in the

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stationary head of the machine. No fracture nor loosening shall occur when a load of 175 pounds is applied for one minute, along the nut axis and toward the front of the instrument.

b. Measurements after test. Perform the dielectric withstanding voltage test specified in 4.7.19.

* 4.7.18 Salt spray. The instruments and mounting hardware shall be tested in accordance with Method 101 of MIL-STD-202 and the following:

a. Test condition. B

b. Salt solution. 20 percent.

c. Measurements after exposure. Visually inspect for corrosion or other evidence or damage to the finish and hardware. There shall be no evidence of corrosion or damage to the finish and hardware.

* 4.7.19 Dielectric withstanding voltage. The instruments shall be tested in accordance with Method 301 of MIL-STD-202 and the following:

a. Special conditions.

(1) Case and input terminals.

(2) Pointer zero adjust control and input terminals.

(3) Input terminals and metal mounting panel.

* 4.7.20 Magnetic shielding. The instrument and a short bar magnet type compass shall be mounted on 7-1/2 inch centers in the same plane and the instrument rotated throughout 360 degrees while observing the effect on the compass. The deflection shall not be greater than 5 degrees.

* 4.7.21 Fungus-inert materials. The instruments shall be tested in accordance with Procedure I, Method 508 of MIL-STD-810. This test need not be performed if the certification specified in Requirement 4 of MIL-STD-454 is provided.

* 4.7.22 Case (seal). This test shall be performed after all other environmental tests. The instruments shall be tested in accordance with Method 112 of MIL-STD-202 and the following:

a. Test condition. B

b. Measurements during test. There shall be no evidence of leakage.

c. Measurements after test. Perform the tests specified in 4.7.2a or b, as applicable, to the instrument type under test.

4.7.23 Operating life. (Six instruments to be tested). The relative humidity need not be controlled during this test. The instruments shall be monitored for evidence of internal moisture (fog or frost).

a. Place the meters in a temperature-humidity test chamber.

b. Adjust the chamber for +25°C ±2°C.

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c. Measure and record the accuracy at 350, 400, and 450 Hz at 115 volts rms or 380, 400, and 420 Hz at 115 volts rms as applicable to the type under test. The accuracy shall be within ±1.5 percent of the applied frequency.

d. Repeat 4.7.11.

e. With the instruments operating at 400 Hz and 115 volts rms, increase the temperature to +71°C. Maintain +69° to +71°C for two hours.

f. During the second hour of the two hour period, turn the instruments ON and OFF (Zero to full scale at a minimum rate of 6 times per minute (360 On/Off cycles)). The maximum rate shall be not greater than 1000 On/Off cycles per hour.

g. With the instruments operating, reduce the temperature to -54°C. Maintain -52 to -54°C for two hours

h. Repeat steps f and e.

i. Repeat steps f, c, and d. Remove from test any instrument that does not meet the accuracy and friction error requirements. When more than one instrument fails to meet the accuracy or friction error requirements, the test shall be terminated.

j. Repeat steps g, f, and e.

k. Repeat steps g and i.

l. Repeat steps b through k until more than one instrument fails to meet the accuracy of friction error requirements, or until the instruments have completed 50,000 ON/OFF cycles, whichever occurs first. Failure of more than one instrument to meet the accuracy of friction error requirements, before achieving 50,000 ON/OFF cycles, shall cause the instruments to fail the life test.

• 4.7.24 Inspection of the preservation, packaging, packing, and marking for shipment and storage. Sample items or packs shall be inspected to determine the preservation, packaging, packing, and marking for shipment and storage conform to the requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 General. The instruments, accessories, and technical publications shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555, Level A, B, and C, as specified (see 6.2) and as specified in 5.2.

5.2 Marking of shipments. The nomenclature for interior packages and external shipping containers shall be as follows:

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Frequency Meter, AC, 3* - 4* Hz; 115v, Type A-*
For 115/200 3 Phase 400 Hz, 4-Wire "Y" System
Type A-*
Specification MIL-F-8378
Manufacturer's Part No. _____**_____

* Applicable type number and frequency range to be entered by the contractor.

** Applicable data to be entered by the contractor.

6. NOTES

6.1 Intended use. The type A-1, A-2, A-3, and A-4 instruments covered by this specification are intended for use in aircraft electrical circuits.

* 6.2 Ordering data. Procurement documents should specify the following:

a. Title number and date of this specification.

b. Selection of applicable level of preservation, packaging, and packing. (See 5.1)

c. Sampling operating life test if required (see 4.4.2a).

* 6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the San Antonio Air Logistics Center, Service Engineering Division, ATTN: MMÆ, Kelly AFB, Texas 78241, and information pertaining to qualification of products may be obtained from that activity.

6.4 Marginal indicia. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 82

Preparing Activity:
Air Force - 82

Project No. 6625-F508

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-8378B	FREQUENCY Meters, Aircraft 380 to 420 Hertz and 350 to 450 Hertz

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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